

Phage therapy for multidrug-resistant *Pseudomonas aeruginosa* infections: a systematic review of the structure and verifiability of the clinical evidence base

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Word count: abstract 246; main text 3 610.

Tables: 4. **Figures:** 2. **Supplementary files:** 6.

Abstract

Background. Multidrug-resistant *Pseudomonas aeruginosa* is a WHO critical-priority pathogen. Several recent syntheses have pooled phage therapy outcomes into overall success proportions. Doing so assumes this evidence base can be aggregated, and that assumption has never been examined.

Objectives. To delimit the clinical literature on phage therapy for resistant *P. aeruginosa* reproducibly, and to determine whether its structure and reporting completeness permit a quantitative synthesis of efficacy.

Methods. Systematic review following PRISMA 2020, with separate streams for databases and trial registers. Nine sources were searched without date restriction, including the regional databases BVS and SciELO. Only reports in English or Spanish were eligible, a criterion adopted after screening closed and verified report by report, including against the full text of every retrieved PDF. The unit of inclusion was the study, not the report.

Results. From 23 057 records, 17 129 unique reports remained. Full-text assessment covered 233 reports, which grouped into 184 studies; 124 of these have a retrievable publication. Single case reports account for 39.5 % and only 18.5 % are comparative designs. Geographic origin is not stated in 61.3 % and resistance class is not stated in 70.2 %. Full text was obtained for 67 studies (54.0 %); the remaining fraction concentrates 56.5 % of the comparative designs. The language criterion removed 35 studies, 18 of them comparative.

Conclusions. This evidence base is simultaneously large and structurally unsuited to a quantitative synthesis of efficacy. Published success proportions rest on assumptions these data do not support.

Keywords: bacteriophages; phage therapy; *Pseudomonas aeruginosa*; multidrug resistance; systematic review; PRISMA.

1. Introduction

Pseudomonas aeruginosa adds to its intrinsic resistance an unusual facility for acquiring new determinants. That combination keeps it among the World Health Organization's priority pathogens ^{1,2}. Once an isolate accumulates resistance to carbapenems, antipseudomonal cephalosporins, fluoroquinolones and aminoglycosides, the clinician's options can be counted on one hand. Sometimes none is both active and tolerable. The problem concentrates in device-associated infections, extensive burns, cystic fibrosis and bronchiectasis.

Phage therapy has re-emerged into that gap. Lytic bacteriophages act through a mechanism unrelated to antibiotic targets, spare the surrounding microbiota, and can be tailored to a patient's own isolate ³. But the clinical literature supporting them did not grow the way evidence for a drug in development grows. It grew as an accumulation of rescue cases published one at a time, almost always after the patient improved.

That growth pattern carries consequences that adding more cases does not dissolve. A published case report is, by construction, an event conditioned on its outcome. A compassionate-use series recruits patients in whom antibiotics have already failed and to whom the treating team judged it reasonable to offer something further ⁴. And when phage is given alongside active antimicrobials, no favourable result can assign the credit

to one or the other⁵.

Published syntheses have nonetheless tended to pool this literature into overall clinical success proportions^{6,7,8}. That step requires taking three things on trust that are rarely checked. First, that the resistance category used for stratification survives comparison with the criteria it claims to apply. Second, that the pooled outcome names the same thing across studies. Third, that the intervention is homogeneous enough for a common proportion to mean anything.

A prior question therefore precedes any estimate, and it is the one this review addresses: does this evidence base have the structure and reporting completeness that a quantitative synthesis requires? Answering it requires first delimiting the literature completely, including the regional literature and the trial registers that previous reviews have covered unevenly, and then characterising what that body of work states and what it leaves unsaid.

2. Methods

2.1 Protocol and registration

This review was conducted following the PRISMA 2020 statement⁹. The completed checklist accompanies the manuscript as supplementary material.

This review is not registered in PROSPERO or OSF. The absence is declared rather than omitted because prospective registration guards against *post hoc* modification of outcomes, and its absence is information readers need in order to calibrate the review. The protocol, the methodological decisions and their dates were recorded immutably in the working repository from the outset (section 2.8).

2.2 Eligibility criteria

Population. Human patients with *P. aeruginosa* infection classified by the authors as multidrug-resistant (MDR), extensively drug-resistant (XDR) or pandrug-resistant (PDR) according to Magiorakos et al.¹⁰, or described in terms implying those categories. Multi-pathogen studies were eligible when the *P. aeruginosa* subgroup was separable.

Intervention. Therapeutic administration of lytic bacteriophages by any route, alone or combined with antimicrobials. Endolysins and other phage-derived products given without the viral particle were excluded as a pharmacologically distinct intervention.

Comparator. None required, given the expected nature of the literature.

Outcomes. Clinical success, microbiological eradication, mortality, adverse events and emergence of phage resistance, as each study defined them.

Designs. Any design with treated patients: randomised and non-randomised trials, cohorts, case series and case reports. Excluded were laboratory, preclinical and modelling work; reviews without primary data of their own; secondary syntheses; veterinary infections; and studies of other organisms without a separable subgroup.

Language. Reports written in English or Spanish were eligible. **This criterion was not part of the original protocol and was adopted on 11 August 2026, with screening already closed**, so the decision was taken in knowledge of what the corpus contained. That is precisely what prospective registration exists to prevent, and it is therefore declared as an amendment, with its impact measured, in section 2.9 and among the limitations.

Language was not inferred from the journal name, which fails in both directions: *Pirogov Russian Journal of Surgery* publishes in English, and an English masthead can carry an article in another tongue. It was taken from the language field declared by Europe PMC, the same datum PubMed files under LA, and, where the source did not index it, from the language field of the export itself, which Scopus, SciELO and BVS all carry. Trial registry entries and Cochrane CENTRAL records were treated as eligible on the published language rules of those sources rather than on assumption.

Verification did not stop at metadata. For the studies with a retrieved full text the language was checked against the **whole article**, analysing the central pages rather than the front matter, which many journals

publish in English even when the body is not. That check exposed three reports whose metadata declared English and whose body is in Russian; all three were excluded. The 233 included reports rest on one of three classes of evidence: analysis of their own text, the language rule of the trial register that publishes them, or the source language field. None is unproven. The assignment for each report, with its evidence class and proof, is supplied as supplementary material.

Unit of inclusion. The **study**, not the report, following PRISMA 2020. A protocol, its conference abstracts and its final publication constitute one study with several reports. The decision and its justification are recorded in the project decision log.

2.3 Information sources and search strategy

Nine sources were searched across the two streams PRISMA 2020 requires to be kept separate. As **bibliographic databases**: PubMed, Scopus in two arms (one organism-first, one intervention-first), Cochrane CENTRAL, BVS including LILACS, CUMED, BINACIS and MedCarib, and SciELO. As **trial registers**: ClinicalTrials.gov, EudraCT and CTIS. No language, date or publication-type restriction was applied. The final run was on 10 August 2026.

Full search strings are supplied as supplementary material, with the literal syntax of each interface, the number of hits and the date each was run. Including BVS and SciELO was a deliberate choice. Clinical experience with phages published in Eastern Europe and Latin America is systematically under-represented in PubMed and Scopus, and leaving it out would have tilted the review towards the anglophone literature without saying so.

2.4 Study selection

Deduplication matched DOI, PMID and NCT transitively using union-find. Identifiers were read **only** from identifier fields, never from free text. The reason is concrete: in a trial run that scanned abstracts, four unrelated articles were merged because all four cited the same earlier paper. Records without an identifier were compared by normalised title and year, with no fuzzy matching.

Selection proceeded in three stages. **Stage 1** applied explicit, auditable rules: no title, publication type without primary data, animal scope declared in MeSH, and phage used as a laboratory tool. **Stage 2** screened titles and **stage 3** screened abstracts. Both used a **closed vocabulary of six exclusion codes** (ORG, REV, SEC, LAB, VET, OFF) fixed before screening began. A reason that fits none of the six does not receive an improvised seventh code: the record advances to the next stage for a human to read.

Decision logs are **append-only**. A correction is a new row superseding the previous one, and both remain. This review had to correct one erroneous exclusion, and that correction is auditable only because the original row survived.

2.5 Positive-control audit

Before screening began, a set of 40 studies known a priori to be eligible was fixed, drawn from an earlier scoping exercise. After each stage we verified that **no study in that set lost all of its reports**. The check operates at study level rather than report level, because excluding an erratum or a duplicate registry entry belonging to an included study is legitimate, whereas losing the study itself is not. The audit passed at all three stages (40 of 40). During one intermediate run it detected a miscalibrated exclusion rule, which was corrected before proceeding.

2.6 Data extraction

The extraction form, with 31 variables and controlled vocabularies, was piloted on five studies before use. Extraction is planned to be **independent and in duplicate** between the two reviewers, with disagreements resolved by consensus and agreement quantified variable by variable (Cohen's kappa).

Status at the time of this report. Duplicate extraction is under way and **has not concluded**. The data reported here come from a systematic pre-extraction from the abstracts of the 124 studies with a retrievable publication, performed with the same variable schema and explicitly flagged as partial in every record. Accordingly, **this report presents no efficacy estimates**. It presents the structure of the evidence base and its reporting completeness, which is what the pre-extraction does support. Outcome variables will be reported once duplicate extraction is complete.

2.7 Risk of bias and certainty

Risk of bias was to be assessed with RoB 2 for randomised trials¹¹, ROBINS-I for non-randomised studies¹² and the Murad domains for case reports and series^{13,14}, and certainty with GRADE¹⁵. These assessments depend on full text and will be reported with the definitive extraction.

2.8 Synthesis methods and reproducibility

The feasibility of a meta-analysis of proportions was assessed explicitly against the usual requirements of that design^{16,17}: interpretable denominators, an outcome comparable across studies, and a sufficiently homogeneous intervention. The result of that assessment is presented in section 3.5.

The entire pipeline is written as versioned, re-runnable code: search, deduplication, screening, grouping and computation of every figure quoted. Each record is keyed on an identifier derived from its content and never on its position in a list, because a positional key changes referent as soon as the set of sources changes. An automated check reconciles the output of each stage against the input of the next, and fails if any declared source is not represented in the corpus. During development that check uncovered an entire database that had been searched and downloaded but never reached the corpus.

2.9 Protocol amendment: the language criterion

On 11 August 2026, with screening complete and 219 studies already delimited, eligibility was restricted to reports written in English or Spanish. The amendment answers an operational limit of the review team, which cannot extract reliably from a Russian text, and is declared here rather than presented as though it had applied from the outset.

Its effect was measured before it was adopted, and it was not small. It excluded **35 reports** and with them **35 whole studies**, none of which retained a report in an admissible language. The corpus fell from 219 to 184 studies and from 159 to 124 studies with a retrievable publication. The loss was not evenly spread: of 41 comparative designs, 23 remained, and of 21 randomised trials, 16.

Three of those 35 were detected only by reading the article itself. Their metadata declared English; the body is in Russian. The third surfaced only on the publisher's own page, which labels the full text as Russian while the abstract is in English. Had verification stopped at the source language field, all three would have stayed in the review.

The positive-control audit also recorded a concrete casualty. **Ronit et al. (2024)**, a case of phage therapy for a *P. aeruginosa*-infected vascular prosthesis published in Danish in *Ugeskrift for Læger*, belonged to the set of 40 studies fixed a priori as eligible and is excluded by the criterion. This is not a screening failure, since the study was identified and classified correctly, but the price of the criterion, and it is reported because an audit that is only reported when it passes demonstrates nothing.

The figures for the review without the amendment survive intact in the decision log, which is append-only, so any reader can reconstruct the earlier corpus and verify the effect independently.

3. Results

3.1 Study selection

The nine sources yielded **23 057 records**, from which 5 928 duplicates were removed to leave **17 129 unique reports** (Figure 1). Stage 1 rules excluded 3 235; of the 13 894 titles screened, 13 434 were excluded and 460 advanced to abstract screening; of those, 227 were excluded and **233 reports proceeded to full text**. Grouped by study, those 233 reports correspond to **184 studies**: 154 with a single report and 30 with several, up to a maximum of eleven reports of one trial.

Of the 184 studies, **124 have a retrievable publication**: 121 articles and 3 that exist only as conference abstracts. The remaining **60 are trial registry entries alone**, with no locatable associated publication. That figure by itself describes a field in which almost one study in three has been registered but has never published results.

Exclusion reasons at title screening were laboratory or preclinical work (4 062), other organism without a separable subgroup (3 031), review or commentary without primary data (2 807), does not evaluate phage therapy in patients (2 634), veterinary scope (838) and secondary synthesis (84). At abstract screening: other organism (65), review or commentary (64), does not evaluate phage therapy (40), **inadmissible language (35)**, laboratory (18) and secondary synthesis (5).

3.2 Full-text retrieval and its bias

Full text was obtained for **67 of the 124 retrievable studies (54.0 %)** through legitimate open-access routes. The remaining 57 require interlibrary loan, institutional access or a request to the authors.

The unretrieved fraction **is not a random sample of the whole** (Table 3). It contains 13 of the 23 comparative studies, **56.5 %**, and reports more patients than the retrieved fraction. Among those unread comparative studies are the field's four landmark randomised trials^{18,19,20,21}; one of them, the Yale trial, has no full journal article, only a conference abstract and results deposited in the register.

The asymmetry is itself a finding. Full-text accessibility covaries with region of publication and with study design, and it pushes in the same direction as the language restriction: both tilt the corpus towards the anglophone open-access literature. A review that declares neither transforms the question without saying so.

3.3 Characteristics of the evidence base

The 124 retrievable studies were published between **2016 and 2026**, and **83.1 %** since 2020. This is a young and rapidly expanding evidence base (Table 1).

Its composition by design is what a rescue literature produces, not what a synthesis of efficacy needs: **49 single case reports (39.5 %)**, 19 case series, 16 randomised trials, 13 prospective cohorts, 7 non-randomised trials and 4 retrospective cohorts. A further 16 do not state a recognisable design in the abstract. Comparative designs total **23 studies, 18.5 %** of the whole, and are the only ones able to sustain a claim of relative efficacy.

Where stated, geographic origin is spread across Russia (7 studies), Poland and Georgia (6 each), Germany (4), France and the United States (3 each), and India, Iran and Israel (2 each). It is not stated in 76 studies. The fall in Russian representation relative to the pre-amendment corpus, from 34 studies to 7, is a direct effect of the language criterion and not of the search.

3.4 What the evidence base does not state

Reporting completeness, measured on what abstracts make explicit, is lowest precisely in the variables on which any stratification depends (Table 2):

- **Resistance class** (MDR, XDR or PDR) is not stated in **70.2 %** of studies.
- **Route of administration** is not stated in **57.3 %**.

- **Pathogen scope**, that is, whether the study covers only *P. aeruginosa* or is mixed with a separable subgroup, is not stated in **52.4 %**.
- Fulfilment of the **DTR** criterion (difficult-to-treat resistance) is not stated in **98.4 %**.

These figures describe reporting in the abstract, not necessarily in the full article, and the distinction matters because some of the information may sit in the body of the text. That said, the abstract is what databases index and what automated reviews consume. Three studies in four failing to state their resistance category there means that no MDR/XDR/PDR stratification built from abstracts can be correct.

3.5 Feasibility of quantitative synthesis

A meta-analysis of proportions requires interpretable denominators, an outcome that names the same thing across studies, and a sufficiently homogeneous intervention. The body of evidence delimited here fails all three requirements.

Denominators. Almost four studies in ten are single cases, in which the success proportion can only take the values 0 or 1 and its binomial variance is maximal. Pooling forty-nine denominators of one with a handful of larger series produces a weighted mean whose weights are set by series size rather than by evidential quality.

Outcome. In this literature "clinical success" ranges from resolution of a fistula at two years to absence of fever at three days. Without a shared definition, the pooled proportion adds numerators that do not count the same events.

Intervention. The products range from fixed-composition commercial polyvalent preparations to cocktails tailored to a patient's own isolate, and in most cases they are co-administered with active antimicrobials. The heterogeneity is conceptual rather than statistical, and no random-effects model resolves it: averaging different interventions yields a number that corresponds to no real intervention.

Added to this is the retrieval bias documented in section 3.2, which as things stand would exclude two-thirds of the comparative studies from any synthesis. **For these reasons no pooled efficacy proportion is presented here**, and we argue that those published in previous reviews rest on assumptions these data do not verify.

4. Discussion

4.1 Principal findings

This review delimits the clinical evidence base for phage therapy in resistant *P. aeruginosa* more broadly than previous syntheses: nine sources, no language restriction, regional literature and trial registers included. The result is 184 studies against the dozens handled by published reviews, and would have been 219 without the language restriction adopted at the end. That increase does not speak well of the field's maturity. Most of it comes from case reports, from registry entries without publication, and from regional literature that searches confined to PubMed and Scopus do not reach.

The central result is negative and worth stating plainly: **the breadth of this evidence base does not translate into capacity for synthesis**. Almost four retrievable studies in ten are a single case. Fewer than one in five is comparative. Seven in ten do not state the resistance category by which stratification is attempted. And almost one identified study in three exists only as a registry entry, with no published results.

4.2 Comparison with previous literature

Recent reviews have reported overall clinical success proportions consistently above 80 %^{6,7,8}. Those results do not contradict ours; they explain them. A literature composed largely of cases published after a favourable outcome yields, when pooled, a high proportion by construction. What that proportion measures is the propensity to publish successes, not the efficacy of the intervention.

Our delimitation further suggests that those syntheses worked on a fraction of the field. The unrestricted search retrieved 34 Russian and 3 Ukrainian studies, and that is precisely where comparative designs concentrate. The language amendment obliges us to set them aside, so this review documents that body of literature and where it sits but cannot speak to its content. That is an explicit debt to the next review with translation capacity.

4.3 Implications

For clinical practice. No published success proportion for phage therapy in resistant *P. aeruginosa* should be used to estimate the probability of benefit for an individual patient. The available evidence supports the statements that phage therapy is being given, that it is tolerated, and that favourable outcomes have been described. It does not support an effect size.

For research. Three shortcomings could be remedied at no added cost and would transform the usefulness of the next reports. State the resistance category together with the explicit criteria used to assign it. Document that the administered phage had verified lytic activity against the patient's isolate. And define clinical success with an explicit time horizon. The first is the most urgent: without it, the population of the question is undefined.

For registries. That 60 of 184 studies are registry entries without a locatable publication is a first-order publication-bias problem, and no statistical method resolves it.

4.4 Strengths and limitations

Strengths. The search covered nine sources across two separate streams, with no language restriction. Selection used a closed vocabulary fixed in advance and was audited against a positive-control set defined before screening began, which allows us to state, rather than merely hope, that screening lost no known eligible study. The whole pipeline is re-runnable code with content-derived keys, and an automated check reconciles each link with the next.

Limitations. There are five, and all are material. **First**, the review is not prospectively registered. **Second**, duplicate extraction has not concluded, so this report presents no outcomes, no risk-of-bias assessment and no GRADE certainty. It is a characterisation of the evidence base, not a synthesis of its results. **Third**, full text was obtained for 54.0 % of the retrievable studies, and the missing fraction is enriched in comparative designs, so what is described here under-represents the comparative part of the field. **Fourth**, screening was performed by a single reviewer supported by explicit rules, with a positive-control audit but without independent duplication. Duplication is planned for the extraction stage. **Fifth**, the restriction to English and Spanish was adopted once screening had closed, and removed 35 studies, 18 of them comparative and one belonging to the positive-control set. It pushes the corpus in the same direction as the retrieval bias and narrows what this review can say about the literature of Eastern Europe, which is where comparative designs concentrate.

5. Conclusions

The clinical evidence base for phage therapy in resistant *P. aeruginosa* is larger than published syntheses reflect and, at the same time, less suited to quantitative synthesis than those syntheses assume. It consists mainly of single cases and non-comparative series, published conditionally on their outcome, with a reporting completeness that does not allow verification of the resistance category defining the population of interest. While those three conditions hold, pooling their outcomes into a common proportion yields a figure that admits no clinical interpretation.

Declarations

Funding. This review received no specific funding from public, commercial or not-for-profit agencies.

Competing interests. The authors declare no competing interests.

Author contributions. [To be completed under CRediT before submission.]

Data and code availability. The screening corpus, the complete decision logs, the extraction form and all pipeline code are available in the project repository. Decision logs are append-only and retain every correction.

Use of artificial intelligence. Generative artificial intelligence tools (Claude, Anthropic) were used in preparing this work to assist with programming the screening pipeline, with pre-extraction from abstracts, and with drafting. The authors reviewed and verified all content and take full responsibility for its accuracy and integrity. No AI tool is listed as an author, in line with ICMJE

recommendations.

Acknowledgements. [To be completed.]

Tables and figures

The four tables are supplied as separate files (paper/tablas/), each with its own footnote. Both figures are supplied as vector PDF (paper/figuras/).

Table 1. Characteristics of the retrievable evidence base (n = 124 studies).

Table 2. Reporting completeness for variables critical to stratification.

Table 3. Comparison of studies with and without full text obtained.

Table 4. Exclusion reasons by stage, using the closed vocabulary.

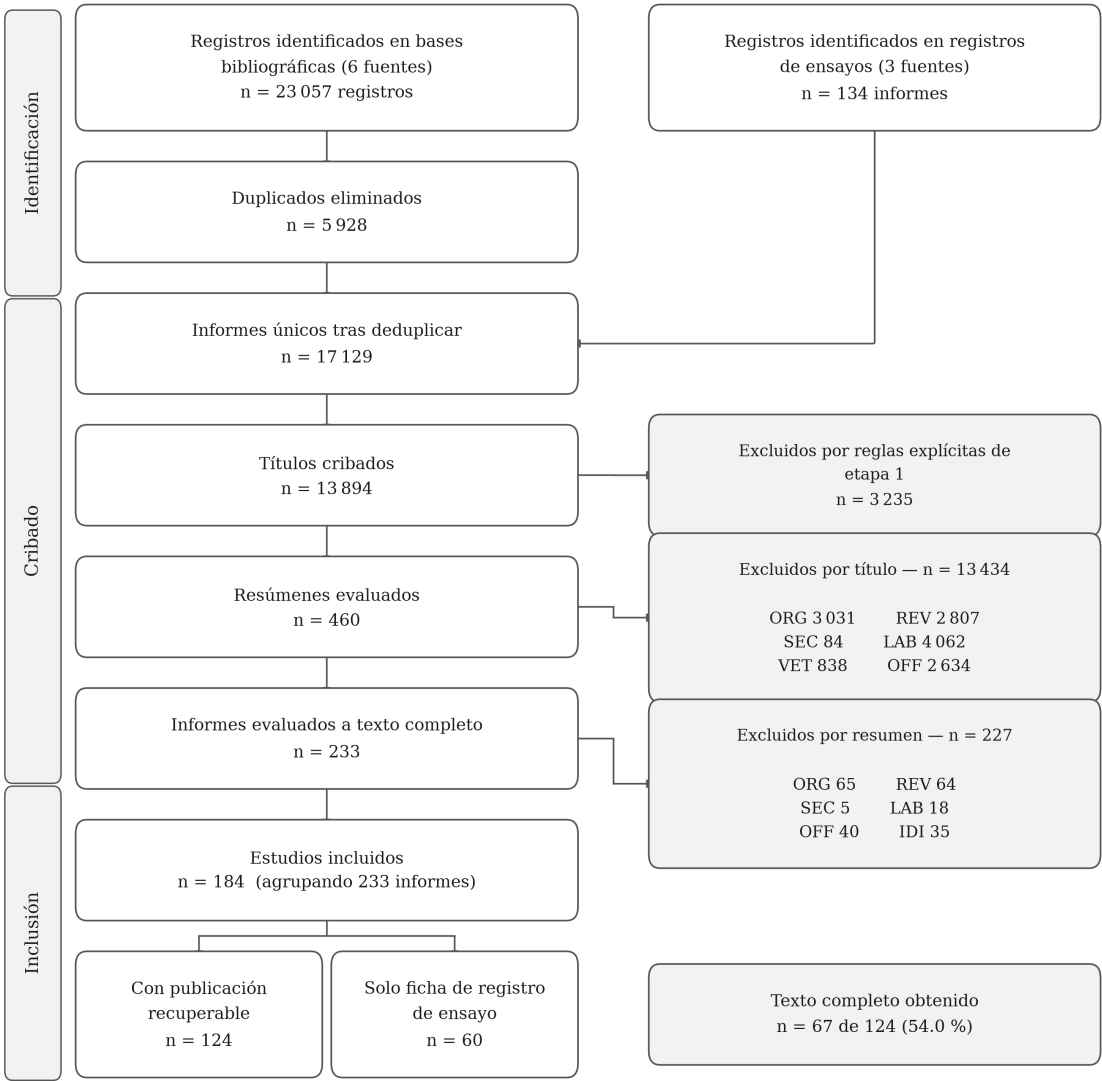


Figure 1. PRISMA 2020 flow diagram.

What it shows. The passage from identified records to included studies, with the two identification streams — bibliographic databases and trial registers — counted separately, as the PRISMA 2020 statement requires. *How to read it.* The left column is the main flow; the grey boxes on the right hold what was excluded at each step, broken down by the closed six-code vocabulary (ORG, REV, SEC, LAB, VET, OFF; defined in Table 4). The register stream enters from the side and does not pass through bibliographic deduplication, because a registry entry has no DOI with which to collide against an article. The final grey box is not part of the selection flow: it reports how many included studies have been read in full text to date. *Source.* Project screening pipeline; final search run, 10 August

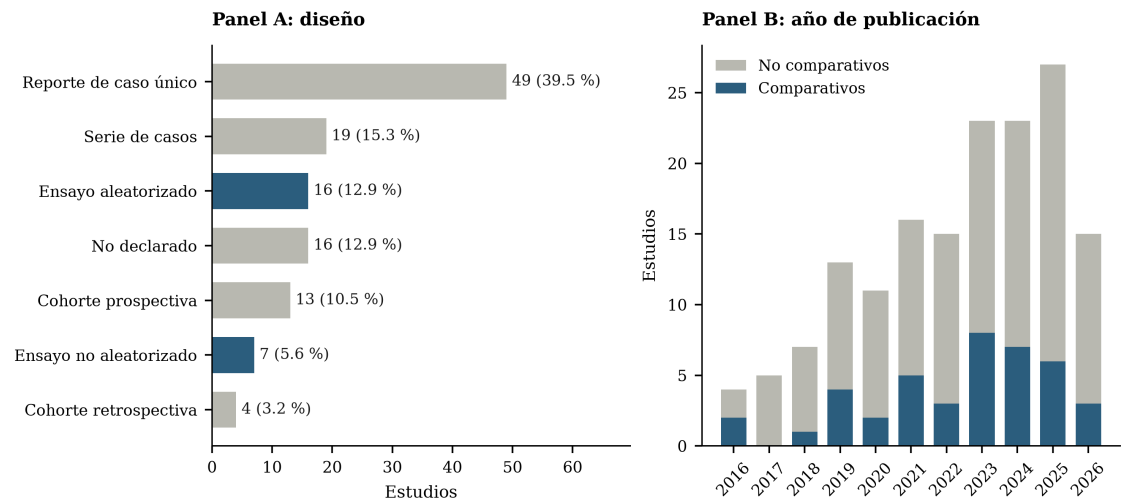


Figure 2. Composition of the retrievable evidence base (n = 124 studies).

What it shows. Panel A orders studies by design; Panel B distributes them by year of publication, separating comparative designs from the rest within each year. *How to read it.* In both panels the dark colour marks comparative designs — randomised and non-randomised trials — the only ones able to sustain a claim of relative efficacy. "Not stated" in Panel A means the abstract does not allow the design to be recognised, not that the study lacks one. The year 2026 is incomplete: the search closed in August. *Source.* Systematic pre-extraction from the abstracts of the 124 studies with a retrievable publication.

Supplementary material

- S1. Completed PRISMA 2020 checklist.
- S2. Literal search strings by database, with date and number of hits.
- S3. Complete screening decision log (stages 1 to 3).
- S4. Extraction form and codebook.
- S5. List of the 219 studies with their status and reports.
- S6. Positive-control audit.
- S9. Language assignment for every report, with evidence class and proof.
- S10. Full-text language verification of every retrieved PDF.

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